

Numerics of the RG-Invariant Entropy-Maximal Locus, $N = 2, 3, 4$

auto-generated by `N4_study/make_summary.py`

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1 Problem and method

Real fully-symmetric quartic couplings $\lambda \in \text{Sym}^4(\mathbb{R}^N)$, $n_{\text{vars}} = \binom{N+3}{4}$. One-loop flow $\dot{\lambda}_{ijkl} = -\lambda_{ijkl} + (\lambda_{ijmn}\lambda_{mnkl} + \lambda_{ikmn}\lambda_{mnjl} + \lambda_{ilmn}\lambda_{mnjk})$; density matrix $\rho_{ij} = \lambda_{iklm}\lambda_{jklm}$; order parameter $T_{ij} = \rho_{ij} - \frac{1}{N}(\text{tr } \rho)\delta_{ij}$. The invariant entropy-maximal locus is the Lie chain

$$T = \mathcal{L}_\beta T = \mathcal{L}_\beta^2 T = \mathcal{L}_\beta^3 T = 0, \quad \mathcal{L}_\beta^k T = k! [t^k]T(\lambda(t)).$$

Solved by multistart Levenberg-Marquardt with Taylor-mode residuals and a complex-step Jacobian. Local manifold $\dim = n_{\text{vars}} - \text{rank } J$ (gap-based rank); moduli = $\dim - \text{orbit dim}$, where $\text{orbit dim} = \text{rank of the } \mathfrak{so}(N) \text{ rotation tangents}$ ($\dim O(N) = \binom{N}{2}$).

2 Master summary

N	n_{vars}	#sols	manifold dim (count)	median	$\max T^{(0-3)} _\Sigma$	Σ -orbit
2	5	196	3 (196)	3	0	196/196
3	15	89	5 (86), 10 (3)	5	1.36×10^{-12}	0/89
4	35	94	9 (27), 11 (18), 14 (49)	14	6.82×10^{-13}	0/94

“ $\max |T|_\Sigma$ ” is the residual of the symmetric ansatz $\lambda_{iiii}=u, \lambda_{iijj}=w$ (it solves the chain at every N). “ Σ -orbit” counts solutions matching the symmetric flattening spectrum $[1, N-1, \frac{N(N-1)}{2}]$.

3 $N = 2$

$n_{\text{vars}} = 5$; 196 solutions, residual $\leq 1.36 \times 10^{-12}$. Single component: manifold dim 3 (196), orbit dim 1 (196) ($\dim O(2) = 1$), moduli 2 (196). Symmetric ansatz residual 0; PCA effective dimension 3.

4 $N = 3$ (agrees with the analytic note)

$n_{\text{vars}} = 15$; 89 solutions, residual $\leq 3.49 \times 10^{-7}$. **Single** component of dimension 5 (86), 10 (3), reconciling with the note’s two-parameter cubic/tetrahedral plane:

$$\underbrace{5}_{\text{embedded}} = \underbrace{2}_{\text{moduli}} + \underbrace{3}_{\dim O(3)}.$$

Spin-2 part vanishes ($q = 0$ in 86/89) and the harmonic part is isotropic ($C = 0$ in 86/89) – exactly the note’s cubic-harmonic condition. Off-symmetric-line solutions give Σ -orbit count 0/89.

5 $N = 4$ (reducible; the $N = 3$ theorem fails)

$n_{\text{vars}} = 35$; 94 solutions, residual $\leq 6.15 \times 10^{-12}$. All are flowing points ($\|\beta\| \in [1.5, 12.3]$, 0 fixed points). The symmetric ansatz still solves the chain (6.82×10^{-13}), but the locus is **reducible**: manifold dim takes values 9 (27), 11 (18), 14 (49).

dim	#sols	orbit	moduli	$q=0$	$\max \ q\ $	iso $C=0$
9	27	6	3	5/27	0.341	5/27
11	18	6	5	0/18	0.479	0/18
14	49	6	8	49/49	0	49/49

Key finding: the spin-2 part $q = \lambda_{ijkk} - \frac{1}{4}(\text{tr})\delta$ is **not** forced to zero – 40/94 solutions have $q \neq 0$ (up to $\|q\| = 0.479$). For $N = 3$ the note proves $q = 0$ necessarily; this is the qualitative break at four flavours. The four-flavour invariant locus is strictly larger than the cubic-harmonic family and contains several inequivalent strata, realising the note’s caveat that higher dimensions admit multiple inequivalent harmonic-isotropy orbits.

6 Reproducibility and caveats

Numbers produced by `N4_study/make_summary.py` (also dumped to `metrics.json`); this PDF interpolates them directly. Solution sets `n2_clean.npz/n3_clean.npz/n4_clean.npz` ($k_{\text{max}} = 3$, 196/89/94 solutions). Caveats: (i) “component” = stratum by local Jacobian rank, not a proven irreducible component; (ii) $N = 4$ convergence is hard (~ 30 – 40% of starts), so rare strata may be under-sampled; (iii) gap-based rank can shift a count by one on near-degenerate spectra.